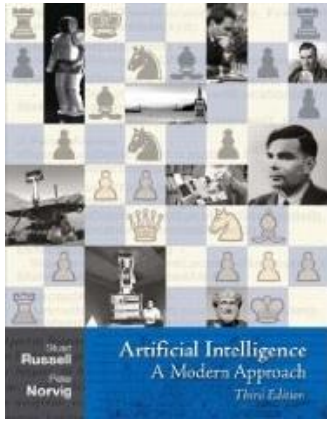
ARTIFICIAL INTELLIGENCE

Topics

- Introduction to AI, History of AI
- AI Domains, AI Applications
- Intelligent Agents and Environment
- Structure of Different type of Agents
- Problem Solving through search
  - (i) Un-informed search (BFS, DFS, Depth First, Depth limited and iterative deepening search)
  - (ii) Informed Search (Greedy best first search, A* search, Heuristics)
  - (iii) Local Search Algorithms (Hill Climbing, Genetic Algorithm)
- Adversarial Search (Minimax Algorithm, Alpha Beta Pruning, Chance Minimax)

  Logical Agents (knowledge-based agents, propositional logic)
- Conditional probability

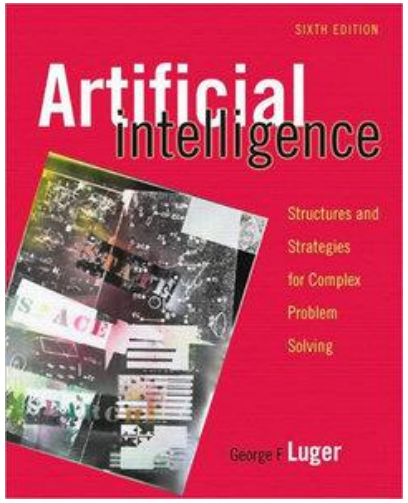
Recommended Texts



Artificial Intelligence : **A Modern Approach**

Third Edition by Stuart J. Russell & Peter Norvig

(2010, Prentice Hall, ISBN 0-13-207148-7) (Course Text)



Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th Edition

George F. Luger

Addison-Wesley (March 7, 2008)

LECTURE 1

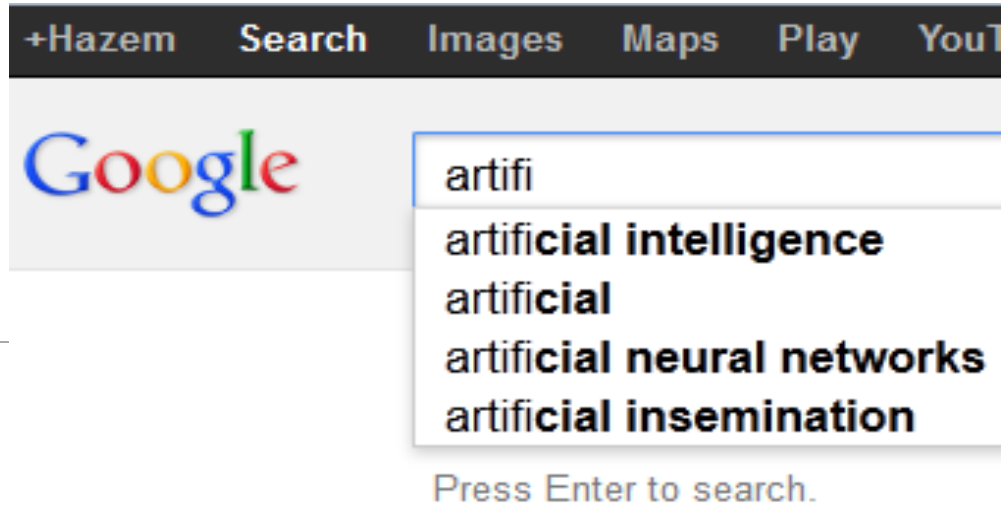
INTRODUCTION TO AI

Where Do We See AI?





Medicine/
Diagnosis



YAHOO!

Search engines

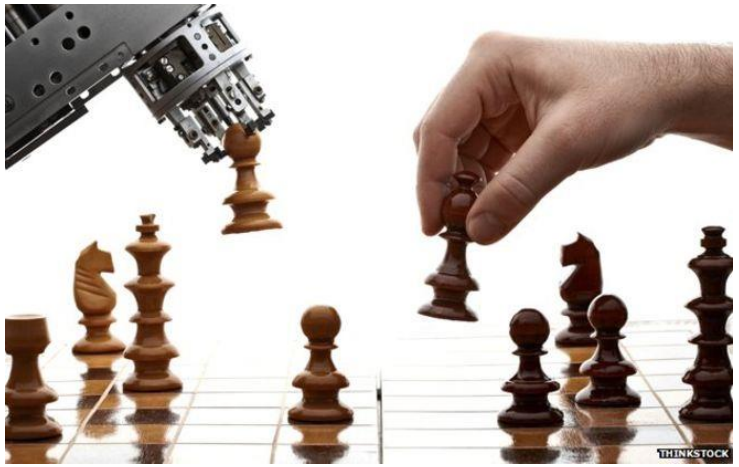


Robotics





Appliances



Games

What is AI?



1. AI is the reproduction of **human reasoning and intelligent behavior** by computational methods.



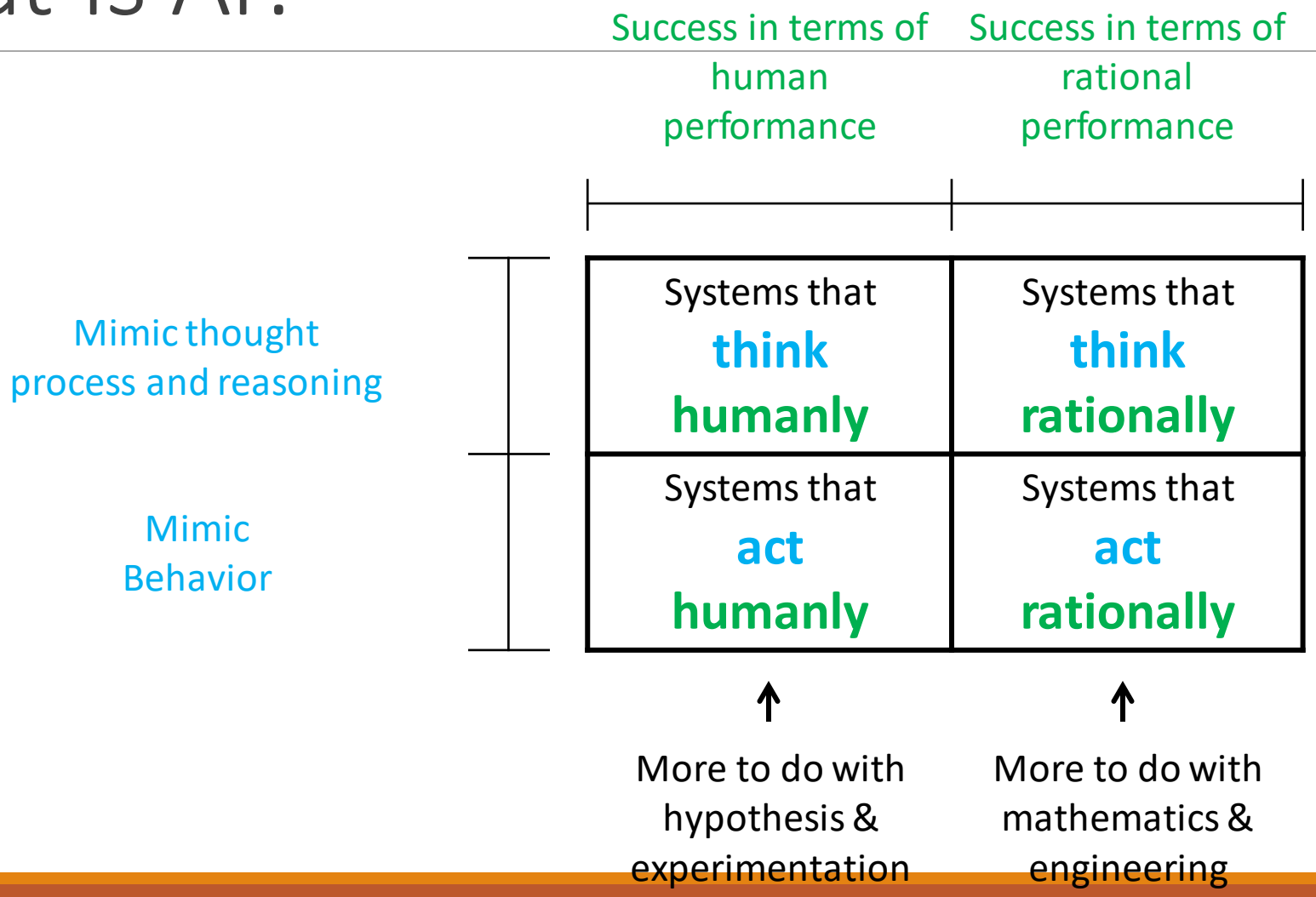
2. AI is *an attempt of* reproduction of **human reasoning and intelligent behavior** by computational methods

What is AI?

The exciting new effort to make computers thinks … *machine with minds,* in the full and literal sense" (Haugeland 1985)	"The study of mental faculties through the use of computational models" (Charniak et al. 1985)
"The art of creating machines that perform functions that require intelligence when performed by people" (Kurzweil, 1990)	A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes" (Schalkol, 1990)

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

What Is AI?


Success in terms of human performance
Success in terms of rational performance
Mimic thought process and reasoning
Mimic Behavior
Systems that think humanly
Systems that think rationally
Systems that act humanly
Systems that act rationally
More to do with hypothesis & experimentation
More to do with mathematics & engineering

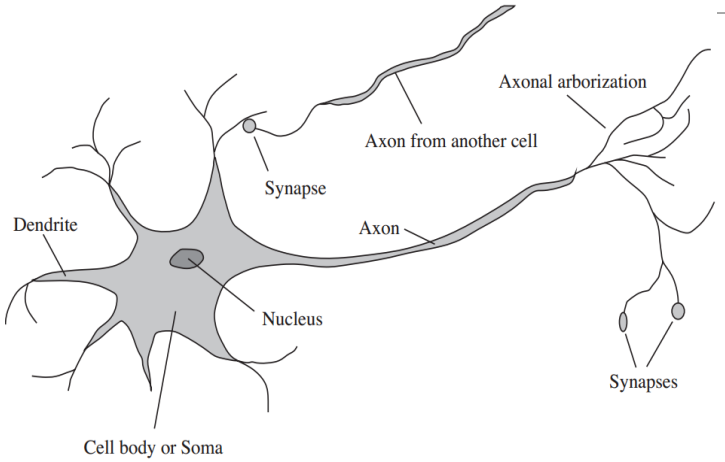


Think Humanly

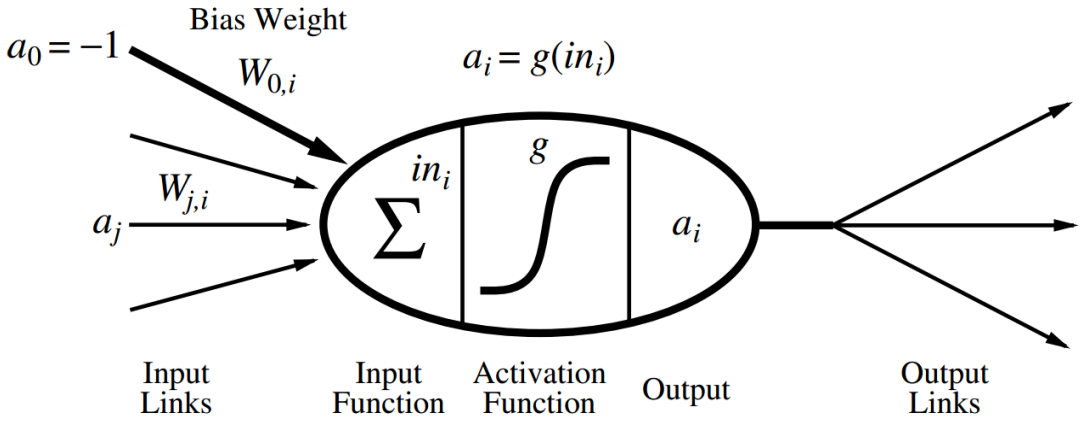
Cognitive modeling approach

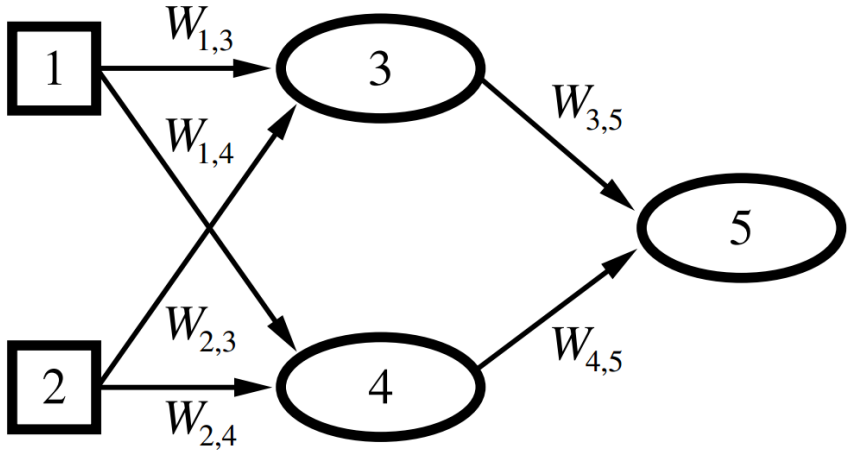
- If we want the computer to think as humans, we must first determine how humans think.
- An interdisciplinary field of **cognitive science**.
- It brings together models from AI and experimental techniques from psychology.
- It tries to come up with precise and testable theories of how the human mind works.
- Example: Neural Networks.

Think Humanly



Neural Network is a set of neuron units.







Think Rationally

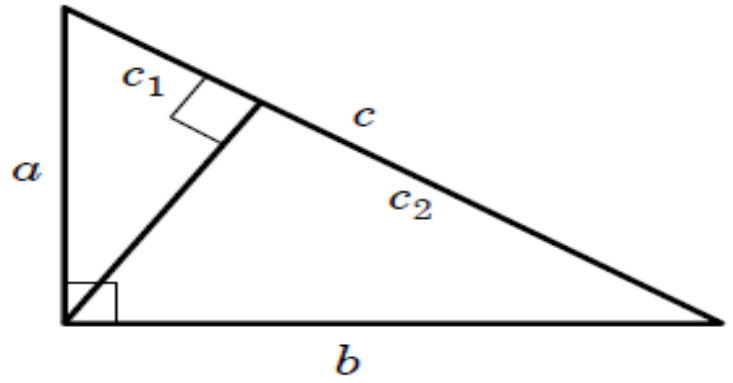
"Laws Of Though" approach:

- Uses logic to reach a decision given some facts via logical inferences.
- Greek philosopher Aristotle was the first to try to codify the way of thinking.
- His deductive reasoning always gave correct conclusions when given correct premises.
- Example: "All Cats are Pets; Bob is a Cat; therefore, Bob is a Pet",
- Example: Theorem proving in first-order logic.

Think Rationally

THINK RATIONALLY: "LAWS OF THOUGH" APPROACH

- Theorem proving:
  - Set of rules:
    - R1: $X = Y / Z \Leftrightarrow X.Z = Y$
    - R2: $X = Y \Leftrightarrow X + Z = Y + Z$
    - R3: $X * Y + X * Z \Leftrightarrow X * (Y + Z)$
  - Set of facts (premises):
    - F1: $b / c \Leftrightarrow c_2 / b$
    - F2: $a / c \Leftrightarrow c_1 / a$
    - F3: $c = c_1 + c_2$
  - Goal:
    - Show that: $c^2 = a^2 + b^2$



Act Humanly

Turing Test Approach





Act Humanly

Turing Test Approach

- To pass Turing test, computer needs:
  - Natural language processing
  - Knowledge representation
  - Automated reasoning
  - Machine learning
- Total Turing Test: interrogator can pass physical objects to the computer (through a hatch).
- To pass Total Turing test, computer needs:
  - Computer vision
  - Robotics



Act Rationally

Rational Agent Approach

- Perceive the environment, and act so as to achieve one's goal.
- Not necessary to do the best action:
  - There is not always an absolutely best action.
  - There is not always time to find the best action.
  - An action that's good enough can be acceptable.
- Example: Game playing.
- Sample approach: Tree-searching strategies.
- Problem: Choosing what to do given the constraints.

Natural vs Machine Intelligence

- Intelligence is not an unique and unshared capability of human. It is more an open collection of attributes than it is a single well-defined entity
- Humans embody many aspects of intelligence while animals typically embody a smaller number of intelligent characteristics, and usually at a much lower level
- Computer based intelligence must be specialized to very restricted domains to be at all comparable to human performance

Human vs Machine Intelligence

- **The two hemispheres of the human brain deal with problems in two distinct paradigms:**
  - sequential (or logical) approach that considers only a small portion of the available data at a time
  - parallel processing looks at data on a global basis

Many tasks which we might reasonably think require **intelligence** are performed by computers **without even thinking** → Complex Arithmetic

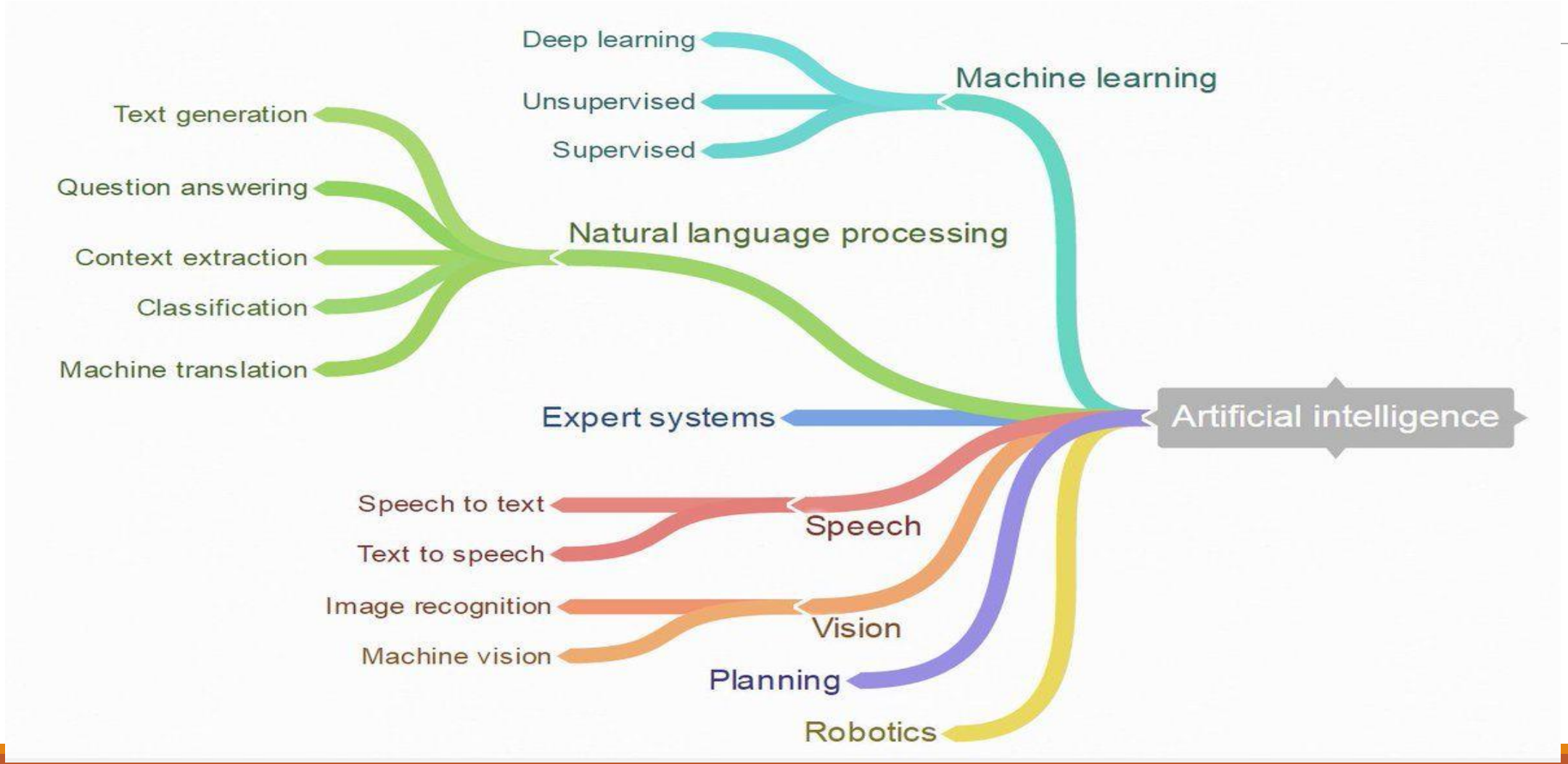
Other tasks that people do without thinking are extremely difficult to automate → Recognizing a Face

Artificial Intelligence

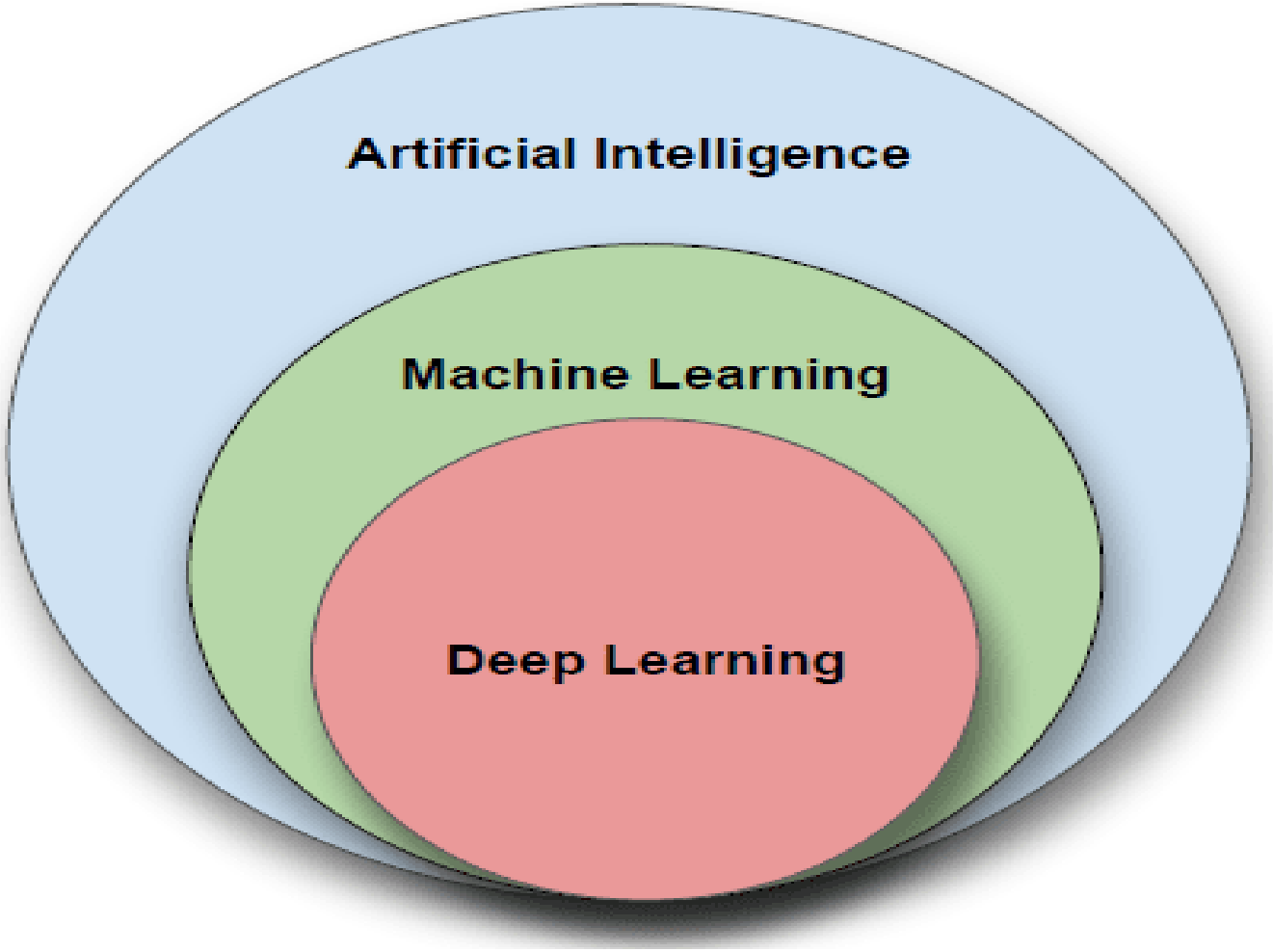
- Artificial intelligence (or AI) is both the intelligence of machines and the branch of computer science which aims to create it, through "the study and design of intelligent agents" or "rational agents", where an intelligent agent is a system that perceives its environment and takes actions which maximize its chances of success.

Where Do We See AI ?! (AI sub-fields)

- **At present, AI has become an interdisciplinary course that involves various fields.**



Relationship of AI, Machine Learning, and Deep Learning



An Application of the Turing Test

CAPTCHA: Completely Automatic Public Turing tests to tell Computers and Humans Apart



What would a computer need to pass the Turing test?

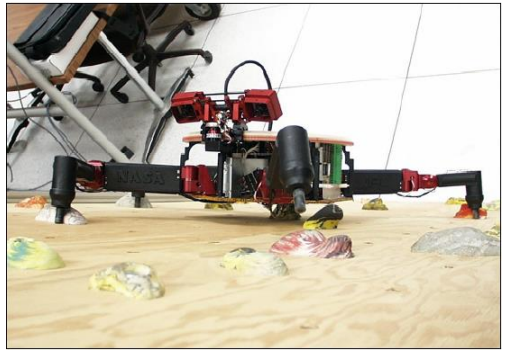
- Natural language processing: to communicate with examiner.
- Knowledge representation: to store and retrieve information provided before or during interrogation.
- Automated reasoning: to use the stored information to answer questions and to draw new conclusions.
- Machine learning: to adapt to new circumstances and to detect and extrapolate patterns.
- Vision (for Total Turing test): to recognize the examiner's actions and various objects presented by the examiner.
- Motor control (total test): to act upon objects as requested.
- Other senses (total test): such as audition, smell, touch, etc.

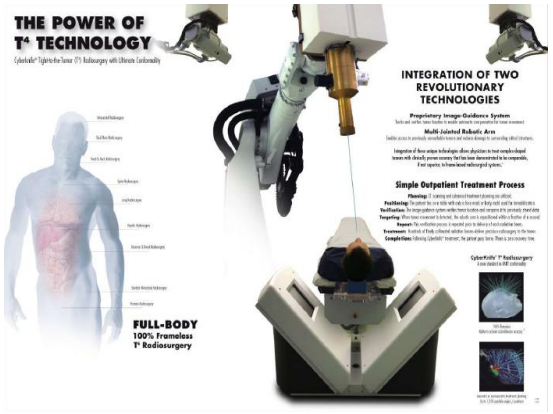
Some Achievements

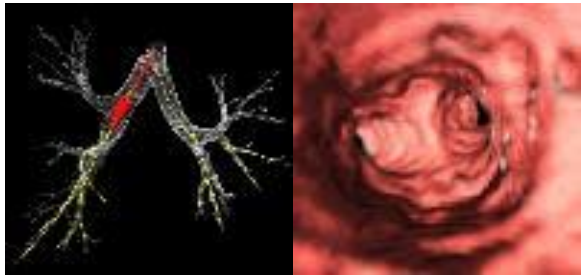
- Computers have won over world champions in several games, including Checkers, Othello, and Chess, but still do not do well in Go
- AI techniques are used in many systems: formal calculus, video games, route planning, logistics planning, pharmaceutical drug design, medical diagnosis, hardware and software trouble-shooting, speech recognition, traffic monitoring, facial recognition, medical image analysis, part inspection, etc...
- DARPA Grand Challenge: robotic car autonomously traversed 132 miles of desert
- Some industries (automobile, electronics) are highly robotized, while other robots perform brain and heart surgery, are rolling on Mars, fly autonomously, ..., but home robots still remain a thing of the future











How to achieve AI?

How is AI research done?

- AI research has both theoretical and experimental sides.
- There are two main lines of research:
  - One is biological, based on the idea that since humans are intelligent, AI should study humans and imitate their psychology or physiology.
  - The other is phenomenal, based on studying and formalizing common sense facts about the world and the problems that the world presents to the achievement of goals.
- The two approaches interact to some extent, and both should eventually succeed. It is a race, but both racers seem to be walking. [**John McCarthy]**

Big Open Questions

- Is intelligent behavior just information processing?
  (Physical symbol system hypothesis)
- If so, can the human brain solve problems that are inherently intractable for computers? Will a general theory of intelligence emerge from neuroscience?
- In a human being, where is the interface between “intelligence” and the rest of “human nature”
  - Self-consciousness, emotions, compulsions
- What is the role of the body?

(Mind-body problem)

- AI contributes to building an information processing model of human beings, just as Biochemistry contributes to building a model of human beings based on bio-molecular interactions
- Both try to explain how a human being operates
- Both also explore ways to avoid human imperfections (in Biochemistry, by engineering new proteins and drug molecules; in AI, by designing rational reasoning methods)
- Both try to produce new useful technologies
- Neither explains (yet?) the true meaning of being human

Bits of History

- 1956: The name “Artificial Intelligence” is coined
- 60’s: Search and games, formal logic and theorem proving
- 70’s: Robotics, perception, knowledge representation, expert systems
- 80’s: More expert systems, AI becomes an industry
- 90’s: Rational agents, probabilistic reasoning, machine learning
- 00’s: Systems integrating many AI methods, machine learning, reasoning under uncertainty, robotics again